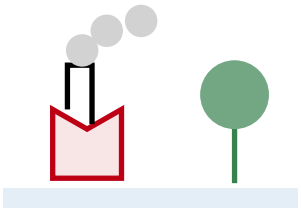
Chapter 10

Externalities



Externalities

Let's consider a paper mill, which uses water to produce paper and may discharge polluted wastewater into a river. The sale of paper benefits the buyer and seller, but the pollution can harm downstream residents who are not part of the transaction.

Definition. An **externality** is an uncompensated effect of an activity on a bystander's well-being.

- A negative externality harms others, as when construction noise interrupts a neighbor's sleep.
- A positive externality benefits others, as when a discovery helps researchers develop new ideas.

Because these effects lie outside private costs and benefits, a trade that benefits its participants need not benefit society as a whole.

Private and Social Costs

To account for a negative production externality, let's distinguish private from social costs. **Marginal private cost** (MPC) is the seller's opportunity cost of producing one more unit.

Marginal external cost (MEC) is the additional harm that this unit imposes on bystanders. Adding it to private cost gives the cost to society:

$$\text{Marginal social cost:} \qquad MSC = MPC + MEC.$$

When pollution increases with output, the social-cost curve lies above the private-cost curve.

Example: Paper Production

Let's put numbers to the paper example in a hypothetical competitive market. We measure Q in tons per day and marginal benefits and costs in NT$1,000 per ton.

Buyers' **marginal benefit** (MB) is their willingness to pay for one more ton. The curves are

$$MB = 120 - Q, \qquad MPC = Q, \qquad MEC = 40.$$

There is no external benefit, so private and social marginal benefits coincide.

- Each ton causes NT$40,000 of harm, so $MSC = Q + 40$.

Why Benefit Falls and Cost Rises

As paper output expands, each extra ton has a lower value to buyers and a higher private cost of production.

- **Marginal benefit falls:** Buyers use the first tons for their most valuable purposes. Additional tons go to less valuable uses, so buyers' willingness to pay falls. Thus, $MB = 120 - Q$ slopes downward.
- **Marginal private cost rises:** Producers use the cheapest production opportunities first. Additional output may require overtime or less efficient equipment, raising the cost of each extra ton. Thus, $MPC = Q$ slopes upward.

Pollution damage is an external cost; it explains the gap between MSC and MPC, not the upward slope of MPC.

MB and MPC as Demand and Supply

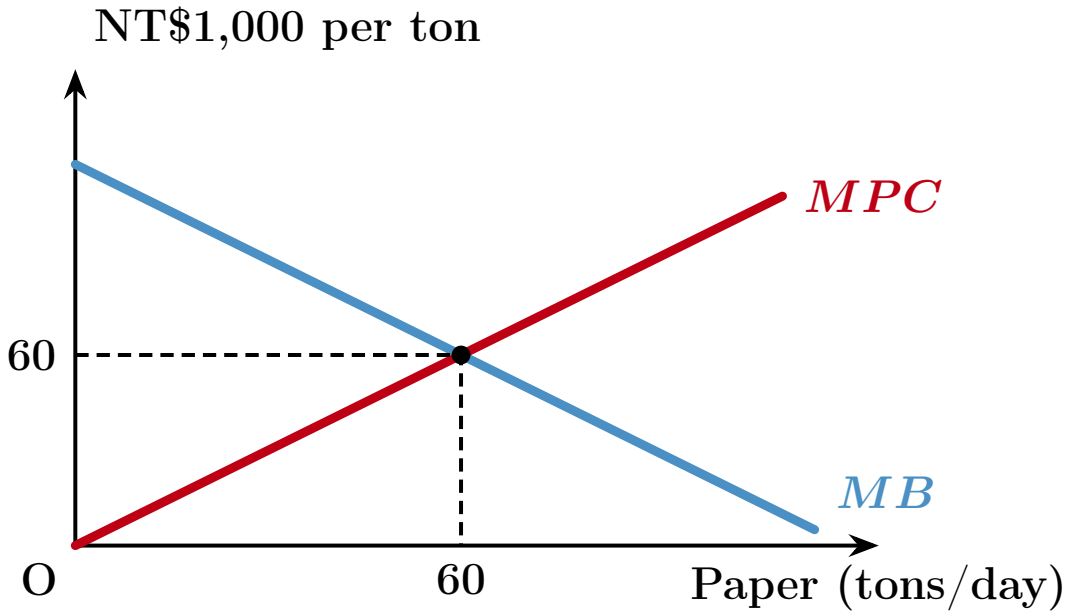
Let's connect these curves to the demand-and-supply framework we already know.

- The MB curve is the **demand curve**: it shows buyers' willingness to pay for one more ton.
- The MPC curve is the competitive **supply curve**: it shows the price needed to induce sellers to produce one more ton.

Without a pollution charge, market equilibrium is where demand meets supply, $MB = MPC$.

The Market Counts Private Costs

Let's find this demand-and-supply intersection. Setting $MB = MPC$ gives $120 - Q = Q$, so the market quantity is $Q = 60$ and the price is $P = 60$.



Social Total Surplus

The market produces 60 tons. Is this best for society?

Definition. Social total surplus is total benefit to society minus total cost to society, including benefits and costs to bystanders.

We will also call this total surplus (TS). Without externalities, taxes, or subsidies, it equals consumer surplus plus producer surplus. External benefits add to this sum, whereas external costs subtract from it.

In the paper market, we count the benefit to buyers, subtract the cost of production, and also subtract pollution damage to bystanders.

Market Equilibrium and Social Welfare

Let's distinguish the market outcome from the outcome that maximizes social total surplus.

- For **market equilibrium**, we use private benefits and costs, including taxes or subsidies, because these guide buyers' and sellers' choices.
- For **welfare**, we use social benefits and costs.

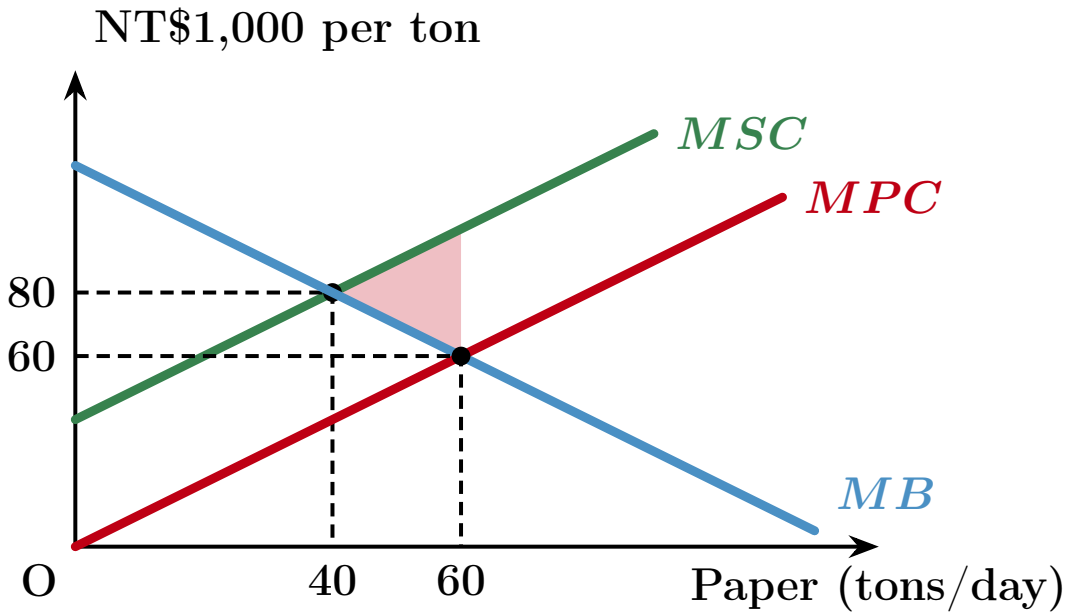
Here there is no external benefit, so MB is also marginal social benefit.

If $MB > MSC$, producing one more ton raises total surplus, whereas if $MB < MSC$, it lowers total surplus. With falling MB and rising MSC, total surplus is therefore highest where $MB = MSC$.

The Efficient Quantity Counts All Costs

Definition. The **efficient quantity**, denoted Q^*, is the quantity that maximizes social total surplus.

In the paper market, Q^* satisfies $MB = MSC$. Solving $120 - Q = Q + 40$ gives $Q^* = 40$, whereas the market produces 60 tons.



Why Reducing Output Can Raise Welfare

Reducing output from 60 tons to the efficient quantity $Q^* = 40$ removes units whose marginal social cost exceeds marginal benefit. This raises total surplus.

The triangle between MSC and MB over tons 40–60 measures the **dead-weight loss** (DWL): social total surplus lost because output exceeds the efficient quantity. Its base is 20 tons and its height is NT$40,000 per ton.

$$DWL = \tfrac{1}{2}(60 - 40) \times 40 = 400.$$

The loss is NT$400,000 per day. The efficient quantity is positive because the first 40 tons are worth their full social cost.

Internalizing the Externality

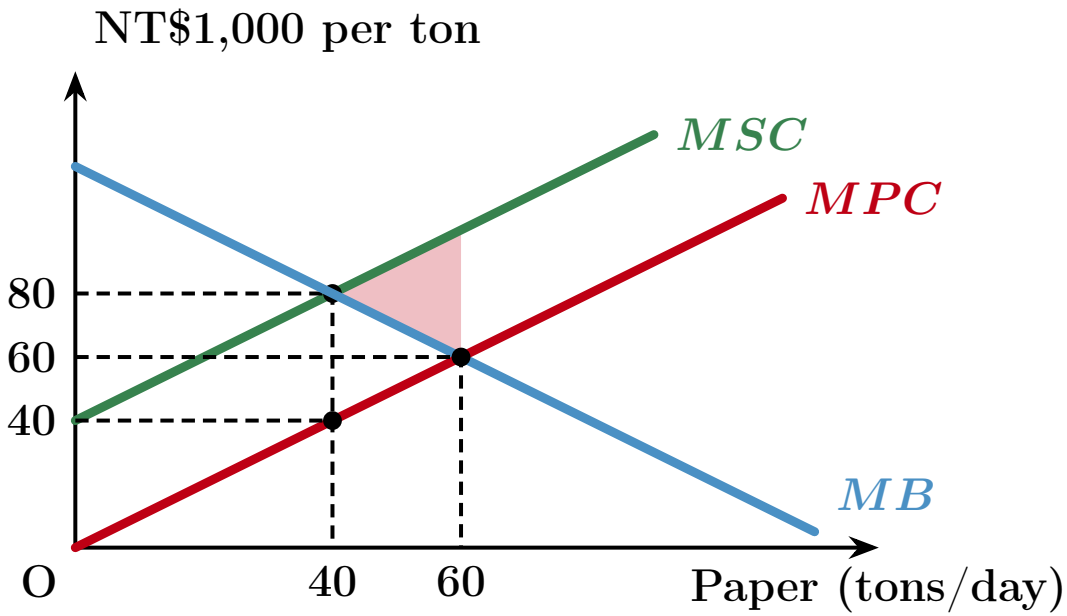
The welfare loss can be avoided by changing incentives so decision makers account for external effects. This is called **internalizing an externality**.

Definition. A **corrective tax**, also called a Pigovian tax, is a tax designed to make decision makers account for the external harm caused by an activity.

In the paper market, a tax of NT$40,000 per ton raises the supply price paid by buyers to $MPC + 40 = MSC$. Producers' marginal private cost is still MPC. The new equilibrium solves $MB = MPC + 40$, giving the efficient quantity $Q^* = 40$. The tax makes private incentives align with social benefits and costs.

Prices after the Corrective Tax

The tax reduces production to $Q = 40$ and creates a gap between the two prices. Buyers pay $P_B = 80$, sellers keep $P_S = 40$, and the tax wedge is $P_B - P_S = 40$, all in NT$1,000 per ton.



Surplus Must Include Pollution Damage

We can also measure social total surplus by adding private surpluses and tax revenue, then subtracting the harm to bystanders:

$$TS = CS + PS + \text{tax revenue} - \text{external damage}.$$

Here CS and PS denote consumer and producer surplus, and TS includes the external damage. The table reports NT$1,000 per day for the paper market.

	No tax	Corrective tax
Consumer surplus	1,800	800
Producer surplus	1,800	800
Tax revenue	0	1,600
External damage	2,400	1,600
Social total surplus	1,200	1,600

Why This Tax Raises Total Surplus

The surplus increase differs from the tax loss in a market without externalities. In that case, a tax prevents some trades whose benefits exceed their costs. That creates deadweight loss.

Here the tax prevents trades whose *social* costs exceed their benefits. The avoided harm exceeds the lost gains from trade.

Tax revenue is a transfer within society. The welfare gain of NT$400,000 comes from changing production, not from collecting revenue.

Positive Externalities

External effects can also create benefits. With a positive consumption externality, demand measures **marginal private benefit** (MPB), whereas **marginal social benefit** (MSB) also includes benefits to others.

$$MSB = MPB + MEB,$$

where MEB is **marginal external benefit**. We now distinguish MPB from MSB, unlike the paper example where both were equal to MB.

- For example, education may help people share useful knowledge with others. The student's higher earnings are a private benefit, whereas benefits to others are the external part.

Example: Training and Shared Knowledge

Let's consider a hypothetical market for training courses. We measure Q in courses per month and marginal values in NT$100 per course.

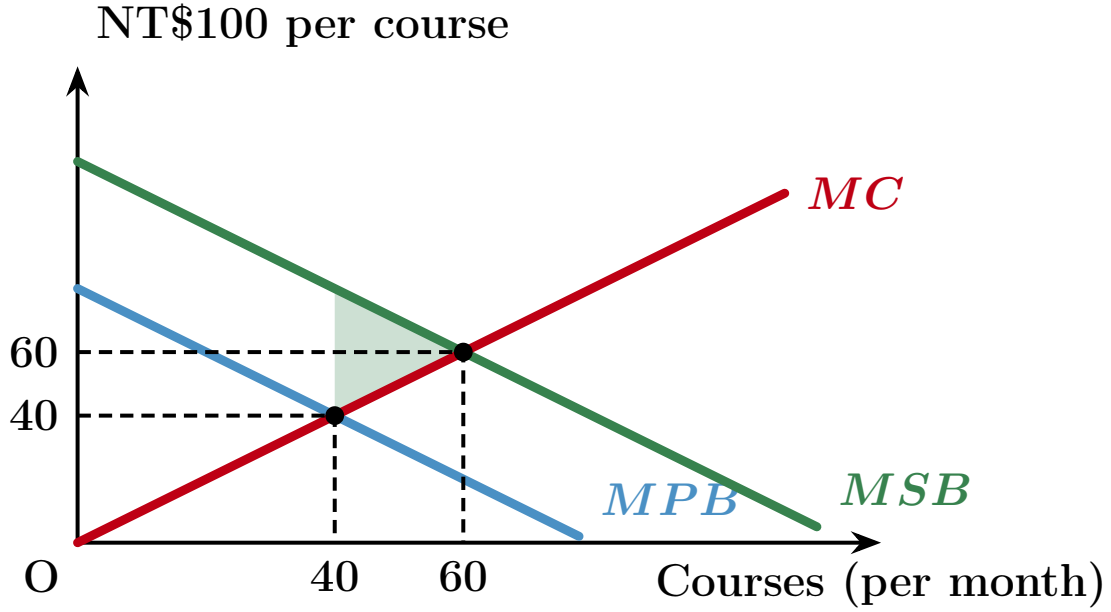
$$MPB = 80 - Q, \qquad MC = Q, \qquad MEB = 40.$$

Students share useful knowledge with coworkers without receiving payment for that benefit. There is no external cost, so MC is both private and social marginal cost.

Adding $MEB = 40$ gives $MSB = 120 - Q$. The market uses $MPB = MC$: $80 - Q = Q$, giving $Q = 40$. The efficient quantity satisfies $MSB = MC$: $120 - Q = Q$, giving $Q^* = 60$.

Too Little of a Beneficial Activity

For courses between 40 and 60, $MSB > MC$, so additional courses raise total surplus. Beyond $Q^* = 60$, $MSB < MC$. The shaded triangle is the gain from expanding enrollment from 40 to 60 courses, or the deadweight loss from underconsumption.



A Corrective Subsidy

To encourage these additional courses, a **corrective subsidy** rewards an activity for its external benefit. In the training example, NT$4,000 per course equals $MEB = 40$ in our units. It raises what providers receive above what students pay, so equilibrium solves

$$MPB + 40 = MC: \qquad 80 - Q + 40 = Q, \qquad Q = Q^* = 60.$$

At $Q = 60$, providers receive NT$6,000 per course. Students pay NT$2,000, and the government pays the NT$4,000 difference.

The shaded gain is $\frac{1}{2}(60 - 40) \times 40 = 400$, or NT$40,000 per month. To recover social total surplus from private surpluses, we subtract government subsidy spending and add the external benefit.

Technology Spillovers and Patents

Like training, research can benefit people who do not pay for it.

Definition. A **technology spillover** is an external benefit of an innovation to other producers who do not fully compensate its creator.

This can weaken the incentive to innovate. Two policies can raise the reward:

- A **research subsidy** uses government funds to lower research costs or increase the inventor's reward.
- A **patent** grants temporary exclusive rights to an invention. Less competition and licensing fees can increase the inventor's earnings and encourage innovation.

Patents act as an **indirect subsidy** financed by users rather than government funds, but can raise prices and restrict access.

Discussion

A student says, "Every tax reduces efficiency, and every activity with a positive externality should receive an unlimited subsidy."

Where does each part of the statement go wrong? Let's distinguish private from social costs and benefits in our answer.

- A corrective tax can improve efficiency by making decision makers account for harm to others. It reduces trades whose social costs exceed their benefits.
- A positive externality calls for more of an activity only up to the point where marginal social benefit equals marginal social cost. A subsidy larger than the marginal external benefit can push quantity beyond that efficient level.

Regulating Pollution Directly

Instead of changing prices through taxes, governments can set pollution limits or require particular production methods.

Definition. Command-and-control policies are regulations that require or forbid particular actions.

Examples include an emissions limit or a required treatment technology.

A binding limit can reduce pollution, but equal reductions at every factory need not achieve a target at the lowest cost.

Reducing pollution is called **abatement**. A factory can do this by treating wastewater, using cleaner equipment, or producing less. The resources used and output forgone are its **abatement costs**. Let's compare these costs across firms.

Example: Two Factories on a River

Let's suppose a paper mill and a tire factory each emit 100 tons of the same pollutant per month. Each ton causes the same harm. The target is 140 tons in total, requiring a reduction of 60 tons.

The mill can cut emissions at NT$3,000 per ton, whereas the tire factory's cost is NT$6,000. These are the firms' abatement costs per ton of pollution removed. The hypothetical costs are constant up to 100 tons per firm.

- Equal cuts of 30 tons cost $30 \times 3{,}000 + 30 \times 6{,}000 =$ NT$270,000.
- Having the mill cut all 60 tons costs $60 \times 3{,}000 =$ NT$180,000.

Concentrating reductions at the mill saves NT$90,000 for the same emissions target. This minimizes the cost of meeting the target; deciding whether the target is efficient also requires comparing abatement costs with avoided harm.

Tradable Pollution Permits

How can we give the factories an incentive to choose these cheaper reductions? The government can issue a limited number of permits and allow firms to trade them.

Definition. A **pollution permit** is a right to emit a specified amount of pollution.

If cutting one ton costs less than a permit, a firm can save money by cutting emissions and selling or avoiding the purchase of a permit. If cutting a ton costs more than a permit, buying the permit is cheaper.

Thus, trading directs reductions toward firms that can cut pollution more cheaply, without changing the cap.

Trading Permits between the Factories

To implement the 140-ton target, the government gives each factory 70 permits for free, each allowing one ton of emissions. Let's suppose the two factories negotiate a sale of 30 permits from the mill to the tire factory at NT$4,500 each.

For each permit traded, the mill receives NT$4,500 for an extra cut costing NT$3,000, whereas the tire factory pays NT$4,500 to avoid a NT$6,000 cut.

- The mill emits 40 tons and cuts 60. Its net cost is $180{,}000 - 135{,}000 =$ NT$45,000.
- The tire factory emits 100 tons and pays NT$135,000 for permits. It makes no reductions.

The combined cost is NT$180,000. Permit payments cancel when we add the firms' costs.

Free Permits Still Have an Opportunity Cost

The permits were given away, but using one prevents its owner from selling it. Thus, even a permit received for free creates an incentive to reduce emissions.

With competitive permit trading and negligible costs of arranging trades, the initial allocation affects who gains from the policy, but not the least-cost allocation of reductions.

So far, the government has given permits to the factories for free. It could instead **auction** them: firms bid to buy permits from the government before trading begins. Auctioning gives the initial permit revenue to the government, whereas giving permits away gives their initial value to recipients.

Taxes Set a Price, Whereas Permits Set a Quantity

We now have two ways to price pollution. An emissions tax fixes the price per ton, and firms choose how much to emit.

A fixed permit cap sets total allowed emissions. Trading determines the permit price when the cap binds.

Does fixing the price rather than the quantity change the outcome? If the government knows firms' abatement costs, it can choose a tax that produces the same emissions as a permit cap.

The distinction matters when costs are uncertain. Under a fixed tax, firms adjust emissions to their costs, so total emissions are uncertain. Under a binding cap, emissions are fixed, whereas the permit price adjusts and is uncertain.

Targeting the Harm Directly

The emissions tax charges for pollution, whereas our paper tax charged for output. The paper example assumes constant pollution damage per ton, so an output tax matches the harm. If firms can pollute less per ton, however, an output tax alone does not reward cleaner production at a given output.

A tax on measured emissions rewards both lower output and cleaner production. Similarly, a fuel tax discourages fuel use, whereas a congestion charge targets driving on a crowded road at a particular time.

The best tax base depends on the external harm and what can be measured and enforced.

Example: Taipei's Designated Garbage Bags

For a local example, Taipei charges for ordinary household waste through designated garbage bags. Separately collected designated recyclables can be handed to collection crews without this charge.

Buying fewer bags lowers a household's payment. Thus, reducing ordinary waste has a private reward, unlike a fixed bill that does not vary with disposal.

Two caveats qualify this example:

- The fee pays for waste services. Its existence alone does not establish that it equals marginal external damage.
- Fewer paid bags need not mean less pollution: households might dump waste illegally instead of reducing it.

Example: A Rush-Hour Trip to Gongguan

The same principle applies to traffic. Let's consider a hypothetical policy of charging for rush-hour car trips to Gongguan. To see how it would work, suppose an extra trip gives its driver NT$120 of benefit and costs NT$80 in fuel, time, and vehicle use.

The trip also delays 20 other travelers by one minute each. Each values time at NT$3 per minute. We assume these are the only external effects.

$$MEC = 20 \times 1 \times 3 = \text{NT\$60}, \qquad MSC = 80 + 60 = \text{NT\$140}.$$

The driver compares $MB = 120$ with $MPC = 80$ and gains NT$40. Society compares $MB = 120$ with $MSC = 140$, so the trip reduces total surplus by NT$20.

Charging for the Delay

Under this hypothetical policy, a NT$60 congestion charge would make the Gongguan driver face the trip's full NT$140 social cost. With a benefit of NT$120, the driver would forgo the trip.

The resulting welfare gain is NT$20. No charge is collected on a trip that does not occur; the gain comes from avoiding a socially costly trip.

For discussion, would switching to an electric car eliminate this externality? It would not eliminate the delay imposed on other travelers. A charge aimed at congestion should depend on where and when the road is crowded.

Private Solutions

Public policy is not the only way to change incentives. People sometimes internalize externalities through private arrangements.

- Social norms can discourage noisy behavior; charitable giving can support activities that benefit others.
- A beekeeper and an orchard owner can contract over pollination, or combine their businesses.
- Neighbors can agree on quiet hours or a payment for noise reduction.

The Coase Theorem

When can such private agreements achieve an efficient outcome?

Definition. An **allocation** specifies how resources are used and who receives goods and services. An allocation is **efficient** if it maximizes social total surplus.

Theorem. *The **Coase theorem** states that private parties can bargain to an efficient allocation when rights are clear and bargaining is costless.*

- The parties can make a transfer whenever a change creates enough total benefit to compensate the person who bears its cost.

With fixed monetary benefits and costs, the initial assignment of rights affects who pays whom, but not which activity maximizes that surplus.

Example: A Noisy Music Practice

Let's suppose a musician values an evening practice at NT$900, but the noise causes NT$1,500 of harm to a neighbor. There are no other effects or bargaining costs.

- If the musician has the right to practice, the neighbor can pay between NT$900 and NT$1,500 to obtain silence. A payment of NT$1,200 makes both better off.
- If the neighbor has the right to a quiet environment, the musician cannot offer enough to compensate the neighbor.

Either way, silence is efficient: stopping practice avoids NT$1,500 of harm at a cost of NT$900 in forgone enjoyment, raising total surplus by NT$600.

Why Bargaining Can Fail

The music example assumes that reaching an agreement costs nothing. If bargaining uses time or other resources, those costs can prevent a deal.

Definition. Transaction costs are the costs of reaching and enforcing an agreement.

In the music example, the gain from stopping practice is only NT$600. Negotiation costs above this gain can prevent a deal when the musician initially has the right to practice.

Bargaining also becomes difficult when many people must agree, rights are unclear, or parties hold out for better terms. A river affected by many polluters and residents illustrates the coordination problem.

Discussion: Cooking Smoke near a Night Market

Let's consider a hypothetical food stall near Gongguan. Installing a filter costs NT$6,000 per month and prevents NT$9,000 of monthly harm to nearby residents. The stall has the right to operate without a filter.

1. If bargaining is costless, what range of payments from residents could induce installation and make both sides better off?

2. Let's suppose reaching and enforcing a deal costs NT$4,000 per month. Is the agreement still worthwhile in total?

We assume the filter leaves food sales unchanged and there are no other effects.

Answer: Bargaining over the Filter

Both sides gain if residents pay more than NT$6,000 but less than NT$9,000 per month. A payment of NT$7,500 gives each side a monthly gain of NT$1,500.

The filter creates NT$3,000 of surplus before transaction costs. Spending NT$4,000 to reach and enforce the deal would exceed that gain.

- One caveat is that residents may each hope their neighbors will pay, making it harder to fund the agreement. A collective rule may help, but its administration and enforcement also use resources.

Takeaways

Efficiency requires counting effects on people outside a transaction.

- Private incentives determine competitive equilibrium, whereas social benefits and costs determine the efficient quantity Q^*.
- Unpriced negative externalities cause excessive activity, whereas unrewarded positive externalities cause too little activity in this model.
- Corrective taxes and subsidies align private incentives with marginal external costs and benefits.
- Tradable permits can achieve a pollution target at lower cost than equal reductions across firms.
- Clear rights and low transaction costs make private bargaining easier.